CAMBRIDGE IGCSE

4 Geometry

0580 PAPER 4 - CALCULATOR

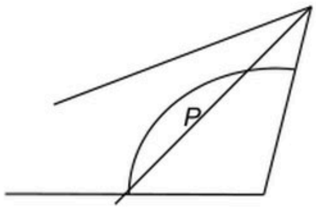
QUESTIONS + MARK SCHEME ANSWERS

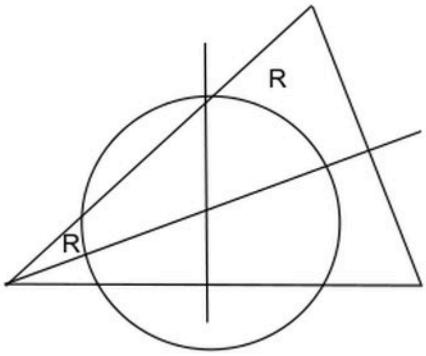
29 complete questions

SUBTOPIC

Constructions, loci and scale drawings

Each question is followed immediately by its official mark scheme answer.

10 **(a)**	475 or 465 to 485	**2**	**B1** for 9.3 to 9.7 [cm] seen
(b)	Correct perpendicular bisector with two pairs of intersecting arcs	**2**	**B1** for accurate with no/wrong arcs or **M1** for correct intersecting arcs
(c)	Compass drawn arc centre *B* radius 5.8	**2**	**M1** for compass drawn arc centre *B* or **B1** for 5.8 cm stated or used
	Accurate angle bisector at *C* with correct intersecting arcs	**2**	**B1** for accurate with no/wrong arcs or **M1** for correct intersecting arcs
		**1**	cao

2	**(a)**	Correct perpendicular bisector of *AB* with 2 pairs of correct arcs isw	**2**	**B1** for accurate with no/wrong arcs or **M1** for correct intersecting arcs with no or wrong line
	(b)	Correct angle bisector at *A* with two pairs of correct arcs isw	**2**	**B1** for accurate with no/wrong arcs or **M1** for two pairs of correct arcs with no or wrong line
	(c)	Circle centre *E* radius 5 cm isw	**2FT**	**FT** circle centre *their E* radius 5 cm provided (a) and (b) attempted **M1** for 250 ÷ 50 oe soi e.g. from arc If 0 scored **SC1** for circle centre *their E*
	(d)		**2**	cao **B1** for each If **0** scored, **SC1** for two 'correct' regions but in part **(c)**, centre correct but radius incorrect

2(a)(i)	9	**1**	
2(a)(ii)	*ABCD* completed accurately with arcs	**2**	**M1** for intersecting arcs radius *their* 9 cm or for *ABCD* completed accurately with no arcs
2(b)	Correct ruled perpendicular bisector of *AB* with 2 correct pairs of arcs Correct ruled bisector of angle *ABC* with 2 correct pairs of arcs Lines intersecting	**4**	**B2** for correct ruled perpendicular bisector of *AB* with 2 correct pairs of arcs or **B1** for correct perpendicular bisector without/wrong arcs and **B2** for correct ruled bisector of angle *ABC* with 2 correct pairs of arcs or **B1** for correct bisector of angle *ABC* without/wrong arcs If lines do not intersect, max **B3**

4(a)	Correct ruled line with *D* marked	**2**	**B1** for correct ruled line or short line
4(b)	47.5	**2**	**B1** for 9.5 or 95 mm seen or for answer figs 465 to figs 485
4(c)	Correct arc radius 7 cm	**2**	**B1** for complete arc other radius, centre *A* or correct but short arc
	Correct ruled perpendicular bisector of *BC* with correct pairs of arcs	**2**	**B1** for correct perpendicular bisector without correct arcs or for correct arcs, no/incorrect line
	Correct ruled bisector of angle *BCD* with correct pairs of arcs	**2**	**B1** for correct angle bisector without correct arcs or for correct arcs, no/incorrect line
	correct region shaded	**1**	Dep on at least **B1B1B1** and five boundaries one of which is an arc
4(d)	[1 :] 500	**1**	

SUBTOPIC

Circle geometry

Each question is followed immediately by its official mark scheme answer.

(ii)	$64\frac{2}{7}$ or 64.3 or 64.28 to 64.29	**1**	
(c) (i)	33 : 20 oe	**2**	**B1** for 33 : 6 or 20 : 6 or 5.5 oe seen or 3.33…oe seen or **M1** for two ratios with a common number of children implied by $20k$ **and** $33k$ seen, $k > 0$
(ii)	236	**3**	**M2** for $\frac{24}{2} \times 11 + \frac{24}{3} \times 10$ oe or $((3 \times 11) + (2 \times 10)) \times 24 \div 6$ or $\frac{6}{6+20+33} \times x \quad 24$ or **M1** for $\frac{24}{2} \times 11$ or $\frac{24}{2} \times 13$ soi or $\frac{24}{3} \times 10$ or $\frac{24}{3} \times 13$ soi oe or $24 \div 6$ soi
(d)	17[.00]	**3**	**M2** for $20.40 \div \left(1 + \frac{20}{100}\right)$ oe or **M1** for (100 + 20)% oe associated with 20.40 seen

(b)	83.6 or 83.60[…]	**2**	**M1** for $\frac{1}{2} \times 15 \times 15 \times \sin(180 - 48)$ oe or $\frac{1}{2} \times 15 \times 15 \times \sin(180 - 2 \times$ *their* **(a)(ii)**$)$ oe
(c)	Opposite angles add up to 180 OR Angle in a semicircle [=90]	**1**	

		6			

4 **(a)** **(i)**	$x \geqslant 5$ oe $y \leqslant 8$ oe $x + y \leqslant 15$ oe $y > x$ oe or $y \geqslant x + 1$	**4**	Condone $5 \leqslant x \leqslant 15$ Condone $0 < y \leqslant 8$ **B1** for each – 1 for first occurrence of strict inequalities used in first 3 inequalities
(ii)	$x = 5$ ruled $y = 8$ ruled $x + y = 15$ ruled $y = x$ ruled broken line	**1** **1** **1** **1**	Allow $y = x + 1$ ruled only after $y \geqslant x + 1$ in **(a)(i)**
	Correct region indicated	**1dep**	Dependent on all marks for lines earned Accept R written in correct quadrilateral or any other unambiguous indication or accept in triangle if $y = x + 1$ used and all marks for lines earned
(b)	78	**2**	**B1** for (7, 8) chosen or **M1** for a calculation shown of the form $6x + 4.5y$ where (x, y) is clearly in *their* region and both x and y are integers
5 **(a)**	37 or [angle] *BAD*	**1**	
	[Angles in] same segment [are equal]	**1dep**	Dependent on 37 or [angle] *BAD*
(b)	74 or 2 [× angle] *BAD* or 2 [× angle] *BED*	**1**	
	Angle at <u>centre</u> is twice angle at <u>circumference</u>	**1dep**	Dependent on 2 × 37 or 2 [× angle] *BAD* or 2 [× angle] *BED* Must use the terms circumference, centre and angle
(c)	143 or 180 – [angle] *BAD* or 180 – [angle] *BED*	**1**	
	[Opposite angles of] cyclic quad [are supplementary]	**1dep**	Dependent on 180 – 37 or 180 – [angle] *BAD* or 180 – [angle] *BED*

8 (a) (i)	Angle A is common to both triangles oe $ADB = ABC$ Third angle of triangles equal oe	1 1dep	Accept $DAB = CAB$ oe Dep on previous mark
(ii)	Similar	1	
(iii)	8.25	2	**M1** for $\frac{16}{12} = \frac{11}{BD}$ oe or better
(b) (i)	38	1	
(ii)	38	1	
(iii)	78	1	
(iv)	26	1	

				Angle $POR = 180 - m$ or $4m$ or $360 - 6m$ Reflex angle $POR = 360 - 4m$ or $6m$ or $180 + m$	

8	**(a)**	[u =] 80 [v =] 160	**1** **1**	
	(b)	6.24 or 6.244 to 6.245	**3**	**M2** for $\sqrt{8^2-5^2}$ oe or **M1** for $l^2+5^2=8^2$ oe or **B1** for suitable right angled triangle drawn with 5 on correct side
	(c)	5.05 or 5.052….	**2**	**M1** for $\frac{4.8}{2.5}=\frac{9.7}{MN}$ oe
	(d)	4 nfww	**4**	**M3** for $[x^n](x+1)=4\times\frac{5}{12}[x^n](x-1)$ oe, n = 1, 2 or 3 or **M2** for $\frac{[x](x+1)}{\frac{5}{12}[x](x-1)}=\left(\frac{2[x]}{[x]}\right)^2$ oe or **M1** for 2^2 or $\left(\frac{1}{2}\right)^2$ soi

2(a)	122	**4**	**B3** for 238 or 61 or 58 correctly identified in working or on diagram or **B2** for 952 seen or 74 or 119 or 29 correctly identified in working or on diagram OR Method 1 using sum of interior angles **M1** for (8 – 2) × 180 or 1080 isw **M1** for *their* 1080 – 4 × 32 **M1** for 360 – *their* 952 ÷ 4 OR Method 2 using isosceles triangles and square **M1** for (180 – 32) ÷ 2 or for 90 **M1** for *their* 74 × 2 + 90 or 90 – *their* 74 **M1** for 360 – *their* 74 × 2 + 90 or 90 + 2(90 – *their* 74) OR Method 3 using four kites joined to centre **M1** for 360 ÷ 4 **M1** for (360 – (*their* 90 + 32)) ÷ 2 **M1** for 2(180 – *their* 119) OR Method 4 using square around outside **M1** for 90 – 32 **M1** for (90 – 32) ÷ 2 **M1** for 180 – 2(*their* 29)
2(b)	105	**3**	**M2** for $360 = 2 \times y + (2y - 60)$ oe or $2(180 - y) = 2y - 60$ oe or **B1** identifying in working or on diagram a relevant angle in terms of y

9(a)	$\frac{5}{8}$ $\frac{3}{8}$ $\frac{1}{6}$ $\frac{5}{6}$ $\frac{7}{10}$ $\frac{3}{10}$	**3**	**B1** for each pair

10(a)	75	**3**	**M2** for 79.5 ÷ 1.06 oe or **M1** for 79.5 associated with 106 [%]
10(b)	962.5 cao	**2**	**B1** for 35 or 27.5 seen
10(c)(i)	16	**1**	
10(c)(ii)	50	**1**	
10(c)(iii)	$\frac{4}{50}$ oe	**2**	**FT** *their* **(c)(ii)** for 1 or 2 marks **B1** for $\frac{4}{k}$, $k > 4$ or $\frac{k}{their 50}$, $k < 50$
10(c)(iv)	19	**1**	
11(a)(i)	12.6 or 12.64 to 12.65	**3**	**M2** for $12^2 + (-4)^2$ OR **B1** for $\begin{pmatrix} 12 \\ -4 \end{pmatrix}$ **M1** for $(their 12)^2 + (their -4)^2$
11(a)(ii)	$\begin{pmatrix} -11 \\ 13 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} -11 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 13 \end{pmatrix}$ or for $[\overrightarrow{BA} =]\begin{pmatrix} -8 \\ 7 \end{pmatrix}$
11(b)	$\frac{1}{2}(\mathbf{b} - \mathbf{a})$ oe	**2**	**M1** for correct route or correct unsimplified answer or **B1** for $\overrightarrow{QS} = \mathbf{b} - \mathbf{a}$ oe
11(c)(i)	$\begin{pmatrix} 9 & 50 \\ 10 & 69 \end{pmatrix}$	**2**	**B1** for 2 correct elements
11(c)(ii)	$\frac{1}{11}\begin{pmatrix} 8 & -5 \\ -1 & 2 \end{pmatrix}$ oe isw	**2**	**B1** for $k\begin{pmatrix} 8 & -5 \\ -1 & 2 \end{pmatrix}$ or $\frac{1}{11}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ or det = 11 soi

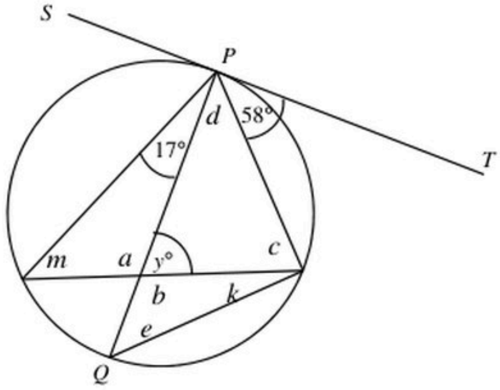
	$[a =]\frac{3}{2}$ oe $[b =]\frac{13}{3}$ oe		e.g. $\frac{1}{6}(2)^3 + 2^2a + 2b$ **A1** for one of e.g. $\frac{1}{6} + a + b = 6$ oe $\frac{8}{6} + 4a + 2b = 16$ oe $\frac{27}{6} + 9a + 3b = 31$ oe $\frac{64}{6} + 16a + 4b = 52$ oe **A1** for another of the above **M1** for correctly eliminating one variable from *their* equations **A1** for $a = \frac{3}{2}$ **A1** for $b = \frac{13}{3}$ oe

	$10m^2 + 9m - 162 = 0$		
9(c)(i)	3.6 and –4.5 final answer	**3**	**B2** for $(2m+9)(5m-18)$ or $\dfrac{-9 \pm \sqrt{(9)^2 - 4(10)(-162)}}{2\times10}$ or better or **B1** for $(am+b)(cm+d)$ where ac = 10 and either $bd = -162$ or $ad + bc = 9$ or for $\sqrt{(9)^2 - 4(10)(-162)}$ or better or $\dfrac{-9 \pm \sqrt{q}}{2(10)}$ or better
9(c)(ii)	20	**1**	

				360	
	10(c)	75.7 to 75.9	**4**	**M1** for $\frac{1}{2}(15+22.4)\times 8$ oe **M2** for $\frac{their(\mathbf{a})}{360}\times\pi\times 8^2$ oe or **M1** for one sector either $\frac{inv\tan\left(\frac{15}{8}\right)}{360}\times\pi\times 8^2$ oe or $\frac{inv\tan\left(\frac{22.4}{8}\right)}{360}\times\pi\times 8^2$ oe	

7(a)	29	**1**	
7(b)	128	**2**	**FT** 180 – 2 (55 – *their* (**a**)) **M1** for angle OCA or angle OAC = 55 – *their* (**a**) soi

2(a)	103	**3**	**M1** for angle *ABC* or angle *ACB* = $\frac{1}{2}(180-26)$ oe **M1** for angle *ABF* = 26 or angle *CBD* or angle *FBE* = 77 or exterior angle *ACB* = 103 correctly identified or in correct position

				or **B1** for 90 soi at (c + k) or between tangent and radius **or** 32 at d **or** 17 at k  	

7(a)	$180-\frac{360}{5}$ or $\frac{(5-2)\times180}{5}$ or $\frac{(2\times5-4)\times90}{5}$ or $\frac{5\times180-360}{5}$	**M2**	or **M1** for $\frac{360}{5}$ or $(5-2)\times180$ or $90(2 \times 5 - 4)$ or $3 \times 180 \div 5$ or $6 \times 90 \div 5$ or $5 \times 180 - 360$ If 0 scored, **SC1** for $\frac{5-2\times180}{5}$
7(b)(i)	7.05 or 7.053…	**3**	**M2** for $12 \times \cos54$ oe or **M1** for implicit form or **B1** for length of edge of pentagon = 14.1 to 14.11 If 0 scored, **SC1** for right angle at M
7(b)(ii)(a)	22.8 or 22.81 to 22.83… nfww	**3**	**M2** for $\frac{\textit{their}\ \textbf{(b)(i)}}{\cos72}$ oe or **M1** for implicit form oe or **B1** for AX = 36.9 or 36.93 to 36.94
7(b)(ii)(b)	179 or 179.1 to 179.3…	**3**	**M2** for $\frac{1}{2}\times12\times \textit{their } AX\times\sin54$ oe or $\frac{1}{2}\times12\times\textit{their } OX\times\sin108$ oe or $\frac{1}{2}\times\textit{their } AX\times\textit{their } OX\times\sin18$ or $\frac{1}{2}\times12^2\times\sin72+\text{area } OBX$ oe or $\frac{1}{2}\times12^2\times\sin72+\text{area } OMB+\text{area } MBX$ oe or **M1** for a correct method to find area of one relevant triangle AOB, OMB, MBX, OBX or ONX **seen**

1(a)	[p =] 132 [q =] 77	3	**B1** for 132 [=p] **B2** for 77 [=q] or **M1** for 180 – (55 + 48) oe or for *their* p – 55
1(b)	74	3	**B2** for $5x - 10 = 360$ or **M1** for $x + (x + 5) + (2x - 25) + (x + 10) = 360$ or for $5x - 10 = k$
1(c)	175	3	**M2** for $180 - \frac{360}{72}$ or for $\frac{180(72-2)}{72}$ or **M1** for $\frac{360}{72}$ or for 180 (72 – 2)
1(d)	[u =] 30 [v =] 60 [w =] 60 [x =] 120 [y =] 40	6	**B1** for 30 **B1** for 60 **B1** for 60 FT *their* v **B1** for 120 FT 2 × *their* w **B2** for 40 or **B1** for angle BDC = 20 or angle ADO = 30 or angle ADB = 70
1(e)	26	4	**B3** for $360 - 22 = 10x + 3x$ oe or better or for $5x + 1.5x = 180 - 11$ oe or better or **M2** for $360 - (3x + 22) = 2 \times 5x$ oe or for $5x + \frac{1}{2}(3x + 22) = 180$ oe or **SC2** for $360 + 22 = 10x + 3x$ oe or better or **M1** for $180 - 5x$, $10x$ or $360 - (3x + 22)$ correctly placed on the diagram or identified or for angle A + angle C = $5x$

8(a)	12	**2**	**M1** for $150 = \frac{(n-2)\times 180}{n}$ or $\frac{360}{180-150}$ oe
8(b)(i)	45	**2**	**B1** for angles at M or K = 45 or angle at L = 90
8(b)(ii)	85	**2**	**B1** for either angle in alt segment = 58
8(b)(iii)	72	**2**	**B1** for either angle at J or H=108 or angle at F=72
8(c)	$OA = OB = OC = OD$ Radii	**B1**	
	$AB = CD$ chords equidistant from centre are equal	**B1**	
	SSS implies congruent	**B1**	

8(a)	[v =] 40 [w =] 80 [x =] 40 [y =] 100 [z =] 60	**5**	**B1** for each **FT** angle z as 140 – *their w*
8(b)	24	**3**	**M2** for $360 - 11x = 2 \times 2x$ oe or **M1** for $360 - 11x$ seen or obtuse angle $KOL = 2 \times 2x$ oe
8(c)(i)	angle ADX = angle BCX oe same segment oe angle DAX = angle CBX oe same segment oe angle $AXD = BXC$ oe [vertically] opposite oe	**M2**	Accept in any order **M1** for one correct pair with reason If 0 scored, **SC1** for two correct pairs of equal angles identified with incorrect/no reasons
	corresponding angles are equal oe	**A1**	
8(c)(ii)(a)	8.75 or 8¾	**2**	**M1** for $\frac{8}{10} = \frac{7}{DX}$ oe
8(c)(ii)(b)	81.8 or 81.78 to 81.79	**4**	**M2** for [cos[BXC] =] $\frac{5^2 + 7^2 - 8^2}{2 \times 5 \times 7}$ oe or **M1** for $8^2 = 5^2 + 7^2 - 2 \times 5 \times 7 \times \cos(...)$ oe **A1** for $\frac{10}{70}$ oe

4(b)(ii)	$\frac{1}{5}$ oe		**M4** for $3\left(\frac{2}{11}\times\frac{1}{10}\times\left[\frac{9}{9}\right]\right)+3\left(\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}\right)$ oe or **M3** for $3\left(\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}\right)$ or **M2** for $3\left(\frac{2}{11}\times\frac{1}{10}\times\left[\frac{9}{9}\right]\right)$ or $\frac{3}{11}\times\frac{2}{10}\times\frac{8}{9}$ oe or **M1** for $\frac{2}{11}\times\frac{1}{10}\times\left[\frac{k}{9}\right]$ where k is 3, 6 or 9
4(b)(iii)	$\frac{131}{165}$ oe	**2**	**M1** for 1 – (*their* **(b)(i)** + *their* **(b)(ii)**) oe
5(a)(i)	81° <u>Angle</u> at <u>centre</u> is <u>twice</u> angle at <u>circumference</u> oe	**2**	**B1** for 81°
5(a)(ii)	81° Alternate segment [theorem] oe	**2**	**FT** *their* **(a)(i)** **B1FT** for 81°

5(a)(iii)	123° Angles on a straight line [= 180] Opposite angles in a cyclic quadrilateral are supplementary oe	3	**FT** *their* acute **(a)(ii)** + 42 **B1** for each element
5(b)(i)	Angle *PTU* = angle *PRQ* corresponding Angle *PUT* = angle *PQR* corresponding Angle *RPQ* is common oe	**M2**	Accept in any order **M1** for one correct pair with reason If 0 scored, **SC1** for two correct pairs of equal angles identified with incorrect/no reasons
	Corresponding angles are equal oe	**A1**	
5(b)(ii)(a)	4 : 7 oe	1	
5(b)(ii)(b)	41.25 oe	3	**M2** for $20\times\left(\frac{7}{4}\right)^2$ oe or $20\times\frac{7^2-4^2}{4^2}$ oe or **M1** for $\left(\frac{7}{4}\right)^2$ or $\left(\frac{4}{7}\right)^2$ or $\frac{7^2-4^2}{4^2}$ or $\frac{4^2}{7^2-4^2}$

3(a)	126 54 117	**3**	**B1** for each
3(b)	angle [in a] semicircle is 90	**B1**	Do not accept triangle for angle
	Allied, co-interior [add to 180] or Angles in triangle [= 180] **and** alternate oe	**B1**	
	32	**B1**	
3(c)	109	**2**	**B1** for 218 or 71 in correct places or correctly labelled

2(a)	52°	3	**M1** for 180 – 2 × 38, implied by 104 **M1** for *their AOB* ÷ 2
2(b)(i)	80°	2	**B1** for *FEC* =50 or *FCE* = 50
2(b)(ii)	100°	1	**FT** 180 – *their* **(i)**

6(a)	15	**2**	**M1** for $\frac{360}{180-156}$ or for $\frac{180(n-2)}{n}=156$ oe
6(b)	38	**2**	**B1** for $AOB = 76$
6(c)	68	**2**	**B1** for $RSP = 68$ or $RQP = 112$

				GOH = *MGK* alternate angles OR Angle *FGM* = angle *GHO* corresponding **and** angle *FGM* = *GKM* alternate segment **and** angle *H* = angle *K* **or M1** for *KMG* = 90, angle in semicircle **or** *OGH* = 90, angle between tangent and radius	
		Two or three pairs of angles equal [so similar] oe	**A1**	Dep on M3 with no incorrect work seen	

2(a)	*PQR* = 90 angle in semi-circle	**B1**	
	PRQ = 61 angle sum of triangle [= 180]	**B1**	
	PSQ = 61 angle in same segment	**B1**	If 0 scored **SC1** for *PSQ* = *PRQ* [= 61] soi
2(b)	57	**4**	**B1** for *ABT* = 98 **B1** for *TAB* or *ATB* = 41 **B1** for *BTC* = 41 or *TBC* = 82 or *ATC* =82 soi

8(a)	$[\cos B =] \ \dfrac{9.5^2 + 7.7^2 - 10^2}{2 \times 9.5 \times 7.7}$ oe	M2	**M1** for $10^2 = 9.5^2 + 7.7^2 - 2 \times 9.5 \times 7.7 \cos B$ oe or better
	70.206 to 70.207 or 70.21 to 70.22	A2	**A1** for $\dfrac{2477}{7315}$ oe or 0.339 or 0.3386….
8(b)(i)	140.4	1	
8(b)(ii)	19.8	1	**FT** (180 – *their* **(b)(i)**) ÷ 2
8(b)(iii)	70.2	1	**FT** 90 – *their* **(b)(ii)**
8(c)	5.31 or 5.314 to 5.315	3	**M2** for $\dfrac{5}{\cos\ their\,\textbf{(b)(ii)}}$ oe or **M1** for $\dfrac{5}{r} = \cos(their\,\textbf{(b)(ii)})$ oe
8(d)	38.8 or 38.9 or 38.78 to 38.85	4	**M3** for $\dfrac{0.5 \times 9.5 \times 7.7 \times \sin 70.2}{\pi \times (their\ \textbf{(c)})^2}$ [×100] OR **M1** for 0.5 × 9.5 × 7.7 × sin70.2 **M1** for $\pi \times (their\,\textbf{(c)})^2$

10(a)	[*DEF*], *BCD* *ADF*, *ADB*	**2**	**B1** for each pair
10(b)	*OQ* *OQT* Tangent perpendicular to radius RHS equal	**5**	**B1** for each

4(a)	144	**2**	**M1** for $180 - \frac{360}{10}$ or $\frac{180(10-2)}{10}$ oe
4(b)	$w = 20$ $x = 20$ $y = 60$ $z = 45$	**5**	**B1** for w **B1FT** for $x =$ *their* w **B2FT** for $y = 80 -$ *their* w or **B1** for angle $BDC = 20$ FT *their* w or angle $ADE = 55$ or angle $CAD = 25$ **B1FT** for $z = 25 +$ *their* w or $105 -$ *their* y

		$[x=]\ 4$, $[y=]\ 5$		or two correct x-values or two correct y-values If B0 scored **and** at least 2 method marks scored, **SC1** for correct substitution of both of *their* x values or *their* y values into $y = x^2 - 4x + 5$ or $y = 2x - 3$	
	4(a)	135	**2**	**M1** for $\frac{360}{5+3} \times k$, where $k = 1$, 3 or 5 oe	
	4(b)(i)	26	**2**	**B1** for $\angle ABD = 49$	
	4(b)(ii)(a)	86	**2**	**B1** for $\angle QAD = 49$ or for $\angle BDA = 45$ or for $\angle BCA = 45$	
	4(b)(ii)(b)	Angle in a semicircle = 90	**1**		
	4(c)(i)	$[2 \times]\cos^{-1}\left(\frac{6.75}{11.5}\right)$ oe	**M2**	**M1** for $\cos(\ldots) = \frac{6.75}{11.5}$ oe	
		108.117…	**A1**		
	4(c)(ii)	12.7 or 12.73 to 12.74	**2**	**M1** for $\frac{108.12}{360} \times 2 \times \pi \times 6.75$	
	4(c)(iii)	100 or 100.1 to 100.2	**3**	**M2** for $\frac{360-108.12}{360} \times \pi \times 6.75^2$ oe or **M1** for $\frac{108.12}{360} \times \pi \times 6.75^2$ If 0 scored, **SC1** for $\frac{360-108.12}{360} \times \pi \times k$	

SUBTOPIC

Similarity and congruence

Each question is followed immediately by its official mark scheme answer.

4	**(a)**	1.6[0] or 1.601 to 1.602	**3**	**M2** for $\frac{0.6}{\cos 68}$ oe or **M1** for $\cos 68 = \frac{0.6}{AC}$
	(b)	43.5 or 43.6 or 43.49 to 43.56	**4**	**M2** for $\frac{1.9^2 + 2.3^2 - their 1.6^2}{2 \times 1.9 \times 2.3}$ or **M1** for implicit statement **A1** for [cos =] 0.724 to 0.726

9(a)	*PMR* = *MSR* = right angle[s] or 90°	**B1**	
	PRM = *MRS* same angle	**B1**	
	AAA oe OR *MPR* = *SMR* 3rd angle of triangle	**B1**	**Dep** on B1B1 and no errors seen

9(b)(i)	5.5	**2**	**M1** for $\frac{x}{4.5} = \frac{9.9}{8.1}$ oe
9(b)(ii)	16.7 or 16.73 to 16.74	**2**	**M1** for $25 \times \left(\frac{8.1}{9.9}\right)^2$ oe or $25 \times \left(\frac{4.5}{\textit{their } 5.5}\right)^2$ oe